

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Professional Portfolio Project Report

1. Executive Summary

This project analyzes customer shopping behavior using a retail transaction dataset containing 3,900 customer purchases. The objective was to identify purchasing trends, customer segments, product performance, subscription behavior, and revenue opportunities. The project demonstrates an end-to-end analytics workflow using Python, PostgreSQL, and Power BI.

2. Business Problem

Retail businesses need data-driven insights to improve customer retention, optimize marketing campaigns, increase revenue, and understand purchasing behavior. This project addresses these challenges through customer and sales analytics.

3. Project Objectives

- Analyze customer demographics and purchasing behavior.
- Identify high-value customer segments.
- Evaluate subscription program effectiveness.
- Discover top-performing products and categories.
- Build an interactive dashboard for decision-making.
- Generate actionable business recommendations.

4. Dataset Overview

Dataset Size: 3,900 transactions

Features Included:

- Age, Gender, Location
- Subscription Status
- Product Category and Item Purchased
- Purchase Amount
- Discounts and Promotions
- Shipping Type
- Review Ratings
- Previous Purchases
- Purchase Frequency

Data Quality:

- Missing values identified in Review Rating.
- Data cleaning and validation performed before analysis.

5. Data Cleaning and Preparation

Python was used for preprocessing and quality improvement.

Key Steps:

- Imported and explored data using Pandas.
- Investigated missing values and data types.
- Imputed missing Review Rating values.
- Standardized column naming conventions.
- Created Age Group segments.
- Engineered Purchase Frequency metrics.
- Removed redundant columns.
- Loaded cleaned dataset into PostgreSQL for SQL analysis.

6. Exploratory Data Analysis (EDA)

The EDA phase focused on identifying trends, distributions, and relationships among variables.

Key Areas Explored:

- Gender-based purchasing behavior
- Category-wise revenue contribution
- Seasonal purchasing patterns
- Review rating distribution
- Subscription impact on spending
- Discount usage patterns

Outcome:

The analysis revealed meaningful customer segments and revenue-driving categories suitable for targeted business strategies.

7. SQL Business Analysis

Ten business questions were solved using PostgreSQL:

1. Revenue by Gender
2. High-Spending Discount Users
3. Top Rated Products
4. Shipping Type Comparison
5. Subscribers vs Non-Subscribers

6. Discount-Dependent Products
7. Customer Segmentation
8. Top Products by Category
9. Repeat Buyers and Subscription Relationship
10. Revenue by Age Group

Each query was designed to answer a practical business problem and support data-driven decision-making.

8. Power BI Dashboard Development

An interactive Power BI dashboard was developed to communicate insights visually.

Dashboard Components:

- KPI Cards
- Revenue Analysis
- Customer Segmentation
- Category Performance
- Subscription Analysis
- Purchase Behavior Trends
- Interactive Filters and Slicers

Business users can explore customer behavior and performance metrics through a user-friendly interface.

9. Key Findings

- Certain product categories generate significantly higher revenue.
- Customer behavior varies across age groups and demographics.
- Subscription status influences customer engagement.
- Repeat buyers contribute strongly to business value.
- Discounts increase purchasing activity but require careful monitoring.
- Product ratings help identify customer preferences and quality perceptions.

10. Business Recommendations

1. Expand customer loyalty programs.
2. Promote subscription-based benefits.
3. Focus marketing on high-value customer segments.
4. Optimize discount strategies to balance revenue and profitability.
5. Highlight top-performing products in promotional campaigns.
6. Use customer segmentation for personalized marketing.

11. Challenges and Solutions

Challenges:

- Missing data values
- Data consistency issues
- Business metric identification

Solutions:

- Data imputation techniques
- Data validation checks
- SQL-driven analytical framework
- Interactive dashboard reporting

12. Skills Demonstrated

Technical Skills:

- Python (Pandas)
- SQL (PostgreSQL)
- Power BI
- Data Cleaning
- Data Transformation
- Feature Engineering
- Data Visualization
- Dashboard Development

Business Skills:

- Customer Analytics
- Revenue Analysis
- Segmentation
- Business Intelligence
- Strategic Recommendations

13. Project Architecture

Raw Dataset (CSV)



Python Data Cleaning & Feature Engineering



PostgreSQL Database



SQL Business Analysis



Power BI Dashboard



Business Insights & Recommendations

14. Future Enhancements

- Predictive sales forecasting.
- Customer churn prediction.
- Recommendation systems.
- Real-time dashboard integration.
- Advanced customer lifetime value analysis.

15. Conclusion

This project demonstrates a complete data analytics workflow from raw data to business insights. By combining Python, SQL, and Power BI, the project transforms transactional data into actionable recommendations that support strategic decision-making. The work showcases practical skills expected from a Data Analyst and serves as a strong portfolio project for internships and entry-level analytics roles.